8.1 Analytical Techniques (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	8. Analysis (A Level Only)
Topic	8.1 Analytical Techniques (A Level Only)
Difficulty	Medium

Time allowed: 100

Score: /83

Percentage: /100

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Question 1a

Verdigris is a green pigment that contains both copper(II) carbonate, $CuCO_3$, and copper(II) hydroxide, $Cu(OH)_2$, in varying amounts.

Both copper compounds react with dilute hydrochloric acid.

$$CuCO_3(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + CO_2(g) + H_2O(l)$$

$$Cu(OH)_2(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + 2H_2O(l)$$

You are to plan an experiment to determine the percentage of copper(II) carbonate in a sample of verdigris. Your method should involve the reaction of verdigris with excess dilute hydrochloric acid.

You are provided with the following:

- 0.494 g of verdigris
- 10.0 mol dm^{-3} hydrochloric acid, HCl (aq)
- commonly available laboratory reagents and equipment.

You may assume that any other material present in verdigris is unaffected by heating and is **not** acidic or basic.

i)
A student suggests that finding the volume of dilute hydrochloric acid required to react with a known mass of verdigris would be a suitable method to determine the percentage of copper(II) carbonate in a sample of verdigris.

Suggest why this method would **not** work.

[1]

ii)
The 10.0 mol dm^{-3} HCl is too concentrated for use in the experiment. Instead, a more dilute solution should be prepared.

Describe how you would accurately prepare 250.0 cm^3 of 0.500 mol dm^{-3} hydrochloric acid from the 10.0 mol dm^{-3} HCl provided.

Your answer should state the name and capacity in cm^3 of any apparatus you would use.

[3]

iii)
The percentage of copper(II) carbonate in a sample of verdigris can be determined by measuring the volume of gas produced when excess hydrochloric acid is added to the sample of verdigris.

Draw a diagram to show how you would set up the apparatus and chemicals to measure the total volume of gas produced in this reaction.

Label your diagram.

[2]

iv)
Sketch a graph on the axes to show how the volume of gas produced would change during your experiment. The independent variable should be on the x-axis.
• Label both axes.
• Extend the graph beyond the point at which the reaction is complete.

[2]

v)
A student thinks that their 0.494 g sample of verdigris only contains $CuCO_3$.

Calculate the minimum volume, in cm^3, of 0.500 mol dm^{-3} HCl that is needed to completely react with this sample if the student is correct.
Show your working.
[M_r: $CuCO_3$ = 123.5]

volume of 0.500 mol dm^{-3} HCl = ... cm^3 [2]

[10 marks]

Question 1b

Azurite is a blue copper-containing mineral. The copper compound in azurite has the formula $Cu_3(CO_3)_2(OH)_2$. This copper compound reacts with sulfuric acid according to the equation.

$$Cu_3(CO_3)_2(OH)_2(s) + 3H_2SO_4(aq) \rightarrow 3CuSO_4(aq) + 2CO_2(g) + 4H_2O(l)$$

A student carries out a series of titrations on 1.50 g samples of solid azurite using 0.400 mol dm^{-3} sulfuric acid.

Assume that any other material present in azurite does not react with sulfuric acid. Some titration data is given in Table 1.1.

Table 1.1

titration	**rough**	**1**	**2**	**3**
final reading / cm^3	25.55	23.90	48.30	28.10
initial reading / cm^3	0.00	0.00	23.90	3.95
titre / cm^3				

The indictor for the titration is bromophenol blue. Bromophenol blue is blue at pH 4.6 and yellow at pH 3.0.

i)
Complete Table 1.1.

[1]

ii)
Calculate the percentage uncertainty in titre 1.

[1]

iii)
The student concludes that 24.15 cm^3 of 0.400 mol dm^{-3} sulfuric acid completely reacts with 1.50 g of azurite.
Calculate the percentage by mass of $Cu_3(CO_3)_2(OH)_2$ in the sample of azurite using the student's value of 24.15 cm^3 of 0.400 mol dm^{-3} sulfuric acid.

Write your answer to **three significant figures.**
Show your working.
[M_r: $Cu_3(CO_3)_2(OH)_2$ = 344.5]

percentage by mass of $Cu_3(CO_3)_2(OH)_2$ in the sample of azurite = ...% [3]

iv)
Identify **two** possible problems with the student's titration experiment and suggest improvements to it.

problem 1 ....

improvement 1 ....

problem 2 ....

improvement 2 ....

[4]

[9 marks]

Question 2a

Activated charcoal is a form of carbon with a very high surface area. It can be used to remove impurities from mixtures. It does this by a process called adsorption, where particles of the impurity bond (adsorb) to the activated charcoal surface.

A student wants to determine the ability of activated charcoal to adsorb a blue dye (the impurity) from aqueous solution.

The equation that links the mass of activated charcoal with the amount of blue dye adsorbed is shown.

$$\log\left(\frac{D}{m}\right) = A + b\log[X]$$

D = difference in concentration of dye (in g dm^{-3}) before and after adsorption
m = mass of activated charcoal (in g)
[X] = final concentration of dye (in g dm^{-3}) after adsorption
A and b are constants

The student uses the following procedure to investigate the ability of activated charcoal to adsorb a blue dye from an aqueous solution.

- Place a 50.0 cm^3 sample of a 25.00 g dm^{-3} solution of blue dye in a conical flask.
- Add a weighed mass of activated charcoal to the flask.
- Stir the contents of the flask for three minutes and then leave for one hour.
- Filter the mixture.
- Determine the final concentration of the blue dye, [X].
- Repeat the procedure using different masses of activated charcoal.

The procedure is carried out. The final concentrations of blue dye, [X], are shown in Table 2.1.

i)
Process the results to complete Table 2.1.

Record your data to **two** decimal places.

Table 2.1

mass of activated charcoal, m / g	initial concentration of blue dye / g dm^{-3}	final concentration of blue dye, [X] / g dm^{-3}	difference in concentration of blue dye, D / g dm^{-3}	$\frac{D}{m}$	$\log\frac{D}{m}$	log [X]
0.20	25.00	0.96		120.20	2.08	
0.25	25.00	0.69		97.24	1.99	
0.30	25.00	0.60		81.33	1.91	
0.35	25.00	0.41		70.26	1.85	
0.40	25.00	0.33		61.68	1.79	
0.45	25.00	0.27		54.96	1.74	

0.50	25.00	0.23		49.54	1.69	
0.55	25.00	0.20		45.09	1.65	
0.60	25.00	0.17		41.38	1.62	

[2]

ii)
Identify the dependent variable in this experiment.

[1]

iii)
State and explain the effect, if any, of increasing the mass of activated charcoal, m, on the amount of adsorption that occurs.

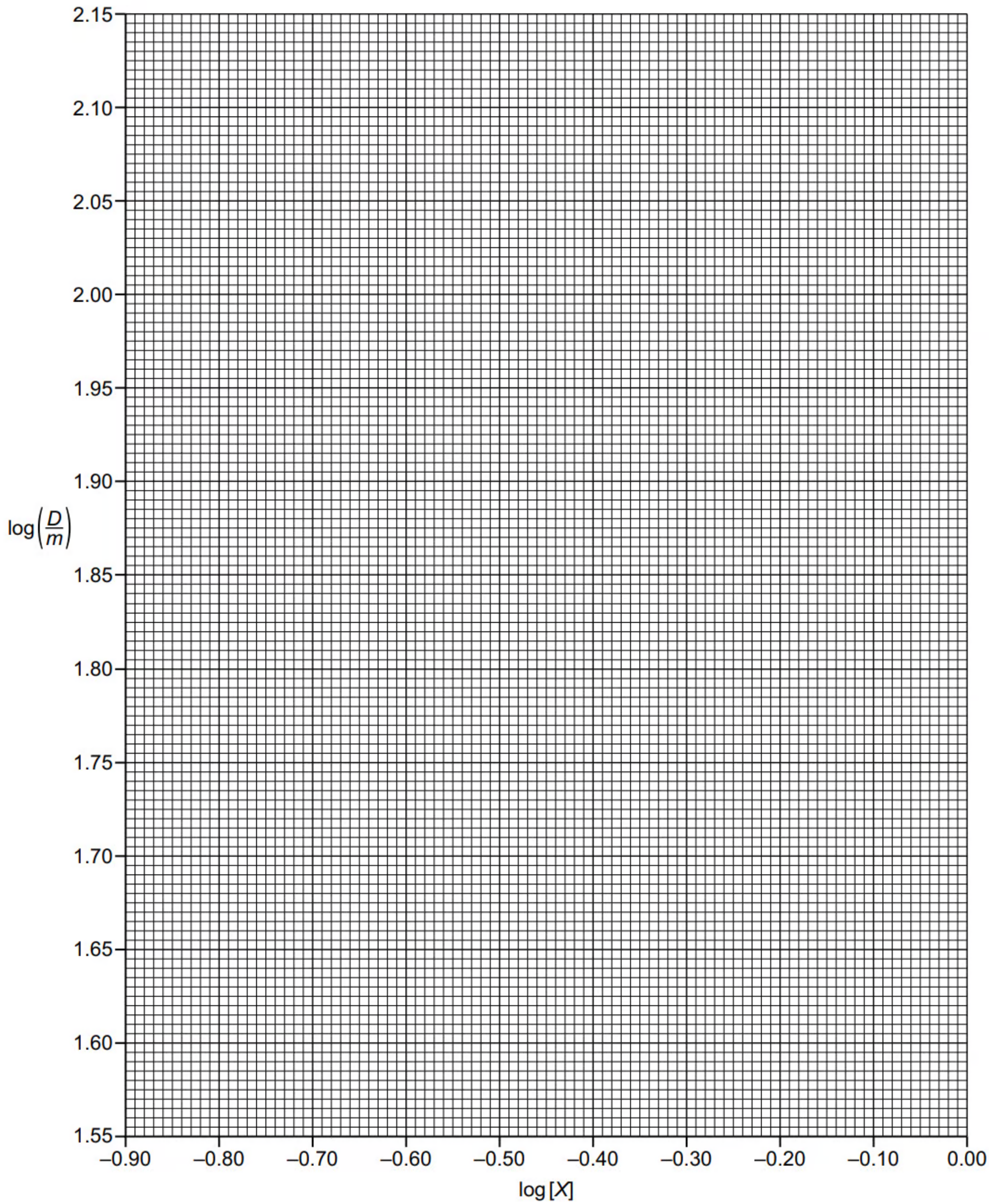
[2]

[5 marks]

Question 2b

Plot a graph on the grid to show the relationship between log $\left(\frac{D}{m}\right)$ and log [X].

Use a cross (×) to plot each data point. Draw the straight line of best fit.


2.15
2.10
2.05
2.00
1.95
1.90
log(D/m)
1.85
1.80
1.75
1.70
1.65
1.60
1.55
–0.90
–0.80
–0.70
–0.60
–0.50
–0.40
–0.30
–0.20
–0.10
0.00
log [X]

[2 marks]

Question 2c

Circle the most anomalous point on the graph.

Suggest why this anomaly may have happened during the experimental procedure.

[1 mark]

Question 2d

i)
Use the graph to determine the gradient of the line of best fit. State the coordinates of both points you used in your calculation. These must be selected from your line of best fit.

Write your answer to **three** significant figures

coordinates 1 .... coordinates 2 ....

gradient = ...

[2]

ii)
Use the graph to determine a value for A.

A = ... [1]

[3 marks]

Question 3a

Column chromatography is a variation of thin-layer chromatography. Column chromatography and gas chromatography work by the same principles of components travelling through the column at different rates.

State the difference between the mobile phases involved in column chromatography and gas-liquid chromatography.

[1 mark]

Question 3b

Three compounds, **A**, **B** and **C,** of similar volatility, are mixed together. The mixture is then analysed in a gas chromatograph.

The gas chromatogram produced is shown in Fig. 3.1.

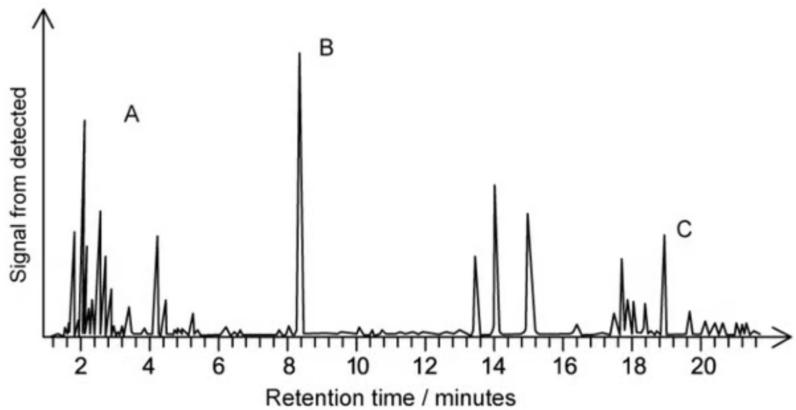


Fig. 3.1

State which compound, **A, B** or **C**, has the greatest affinity for the solid phase. Explain your reasoning.

[2 marks]

Question 3c

Use the gas chromatogram shown in Fig 3.1 to identify the most abundant compound in the mixture. Explain your answer.

[1 mark]

Question 3d

An oil tanker is travelling through the English Channel. The tanker has a slight leak which is not large enough to result in an oil slick but some oil is noticed on a beach.

Suggest how gas chromatography could be used to identify the tanker as the source of crude oil pollution on the beach.

[2 marks]

Question 4a

A mixture is analysed using thin-layer chromatography.

Draw the labelled apparatus that would be used to identify the number of compounds in the mixture.

[2 marks]

Question 4b

A student runs a thin-layer chromatography experiment and plans to determine the compounds from their R_f values.

Describe the steps that the student needs to perform to determine the identity of the compounds.

[4 marks]

Question 4c

The student's results are shown in Fig. 4.1.

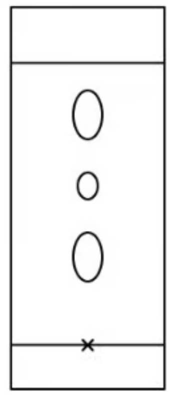


Fig. 4.1

For their measurements, the student locates the centre of each spot by estimating its rough position by eye.

Suggest an improved method to locate the centre of each spot.

[2 marks]

Question 4d

The student runs another TLC practical on a mixture of benzaldehyde and benzyl alcohol using 7:3 pentane / diethyl ether as a solvent.

The student's chromatogram is shown in Fig. 4.2.

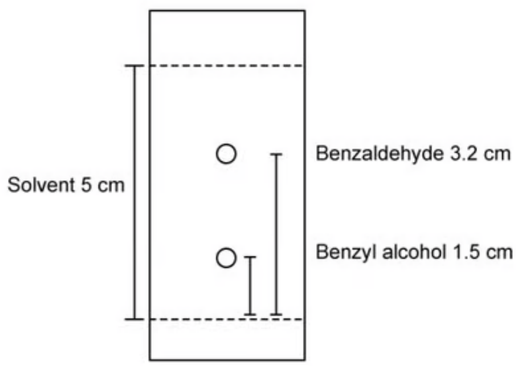


Fig. 4.2

Calculate the R_f values for both compounds in the chromatogram.

[2 marks]

Question 4e

Explain why the maximum R_f value of a compound cannot exceed 1.

[2 marks]

Question 5a

During the production of an NMR spectrum, tetramethylsilane (TMS) is mixed with the sample.

i)

Give the structural formula of the standard reference chemical used for ^{1}H NMR spectroscopy.

[1]

ii)

Explain why tetramethylsilane is used as the standard reference chemical.

[2]

[3 marks]

Question 5b

State the number of peaks in the C-13 NMR spectrum of 1,3-dichlorobenzene.

[1 mark]

Question 5c

i)

Predict the number of peaks in the ^{13}C NMR spectrum of ethylbenzene, shown in Fig. 5.1.

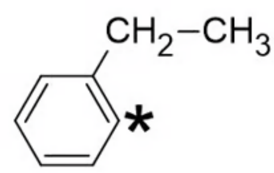


Fig. 5.1

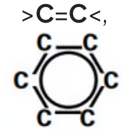
[1]

ii)

The data in Table 5.1 should be used in answering this question.

One of the carbon atoms in the structure of ethylbenzene shown in Fig. 5.1 is labelled with an asterisk (*). Suggest a C-13 chemical shift range for this carbon environment.

Table 5.1

Hybridisation of the carbon atom	Environment of carbon atom	Example	Chemical shift range δ/ppm
sp^3	alkyl	$\mathbf{C}H_3$–, $\mathbf{C}H_2$–, –**C**H<, >**C**<	0 – 50
sp^3	next to alkene / arene	–**C**–C=C, –**C**–Ar	25 – 50
sp^3	next to carbonyl / carboxyl	**C**–COR, **C**–O_2R	30 – 65
sp^3	next to halogen	**C**–X	30 – 60
sp^3	next to oxygen	**C**–O	50 – 70
sp^2	alkene or arene	>**C**=**C**<, 	110 – 160
sp^2	carboxyl	R–**C**OOH, R–**C**OOR	160 – 185
sp^2	carbonyl	R–**C**HO, R–**C**O–R	190 – 220
sp	nitrile	R–**C**≡N	100 – 125

[1]

[2 marks]

Question 5d

Compound **A** contains the elements carbon, hydrogen, oxygen and nitrogen only. Compound **A** contains a benzene ring.

Part of the mass spectrum of **A** is shown in Fig. 5.2.

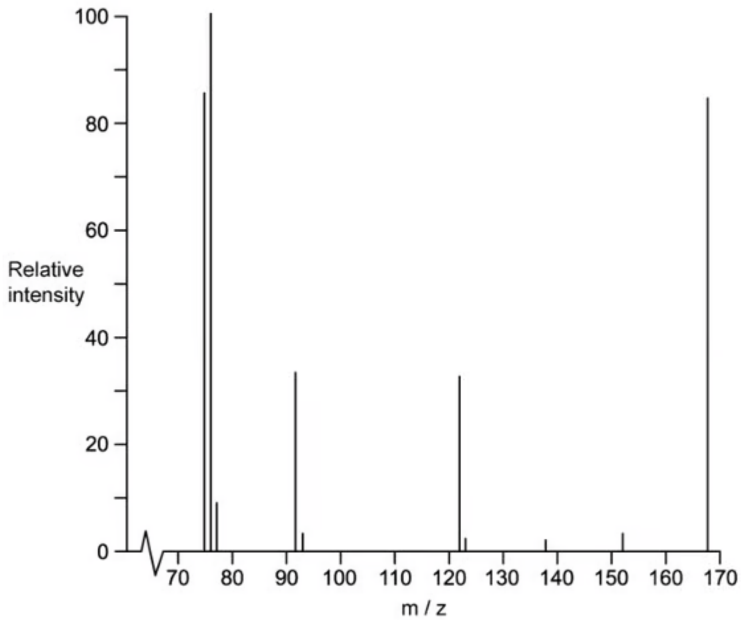


Fig. 5.2

i)

Give the identity of the molecular ion that gives rise to the peak at *m / e* = 76 in the Fig. 5.2.

[1]

ii)

Suggest the structures of the three possible dinitrobenzene isomers of **A** that contain a benzene ring.

[3]

iii)

The C-13 NMR spectrum of compound **A** has four peaks. Identify the structure of **A**. Explain your reasoning by labelling the different carbon environments in **all** the structures drawn in part **(ii)**.

[3]

[7 marks]

Question 6a

Methyl cinnamate, $C_{10}H_{10}O_2$, is a white crystalline solid used in the perfume industry.

The proton NMR spectrum of methyl cinnamate in the solvent $CDCl_3$ is shown in Fig. 6.1.

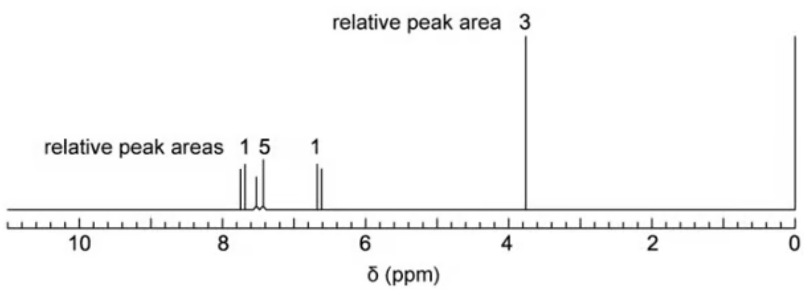


Fig. 6.1

i)

Explain why $CDCl_3$ is used as a solvent instead of $CHCl_3$.

[1]

ii)

Explain why TMS is added to give the small peak at chemical shift $\delta = 0$.

[1]

[2 marks]

Question 6b

The structure of methyl cinnamate is shown in Fig. 6.2.

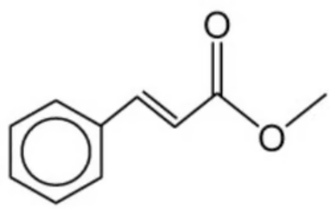


Fig. 6.2

The data in Table 6.1 should be used in answering this question.

Identify the proton environment that gives rise to the peak at a chemical shift of 3.8 ppm in Fig. 6.1. Explain your answer.

Table 6.1

Environment of proton	Example	chemcial shift range, δ / ppm
alkane	$-CH_3$, $-CH_2-$, $>CH-$	0.9 – 1.7
alkyl next to C=O	$CH_3-C=O$, $-CH_2-C=O$, $>CH-C=O$	2.2 – 3.0
alkyl next to aromatic ring	CH_3-Ar, $-CH_2-Ar$, $>CH-Ar$	2.3 – 3.0
alkyl next to electronegative atom	CH_3-O, $-CH_2-O$, $-CH_2-Cl$	3.2 – 4.0
attached to alkene	$=CHR$	4.5 – 6.0
attached to aromatic ring	$H-Ar$	6.0 – 9.0
aldehyde	$HCOR$	9.3 – 10.5
alcohol	ROH	0.5 – 6.0
phenol	$Ar-OH$	4.5 – 7.0
carboxylic acid	$RCOOH$	9.0 – 13.0
alkyl amine	$R-NH-$	1.0 – 5.0
aryl amine	$Ar-NH_2$	3.0 – 6.0
amide	$RCONHR$	5.0 – 12.0

[3 marks]

Question 6c

Proton NMR spectroscopy can be used to distinguish between isomers of $C_6H_{12}O_2$.

Draw the two esters with formula $C_6H_{12}O_2$ that each have only two peaks, both singlets, in their ^1H NMR spectra. The relative peak areas are 3:1 for both esters.

[2 marks]

Question 6d

The proton NMR spectrum of another isomer of $C_6H_{12}O_2$ is shown in Fig. 6.3.

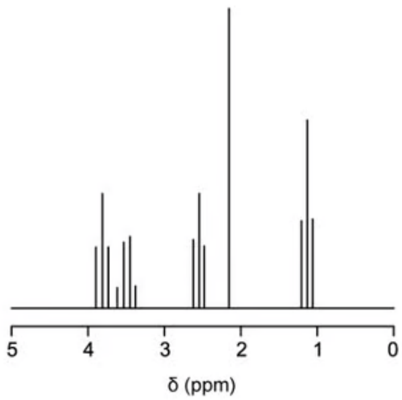


Fig. 6.3

The integration values for the peaks in the proton NMR spectrum of this isomer are given in Table 6.2.

Table 6.2

Chemical shift, δ/ppm	3.8	3.5	2.6	2.2	1.2
Integration value	0.6	0.6	0.6	0.9	0.9
Splitting pattern	triplet	quartet	triplet	singlet	triplet

i)
Deduce the simplest ratio of the relative numbers of protons in each environment in the isomer.

[1]

ii)
The data in Table 6.1 should be used in answering this question.

Describe and explain the splitting patterns of the peaks at $\delta = 3.5$ and $\delta = 1.2$.

splitting pattern at $\delta = 3.5$ ....

reason for splitting pattern at $\delta = 3.5$ ....

splitting pattern at $\delta = 1.2$ ....

reason for splitting pattern at $\delta = 1.2$ ....

[4]

[5 marks]

Question 6e

Four isomers of $C_6H_{12}O_2$, **A**, **B**, **C** and **D**, are shown in Fig. 6.4.

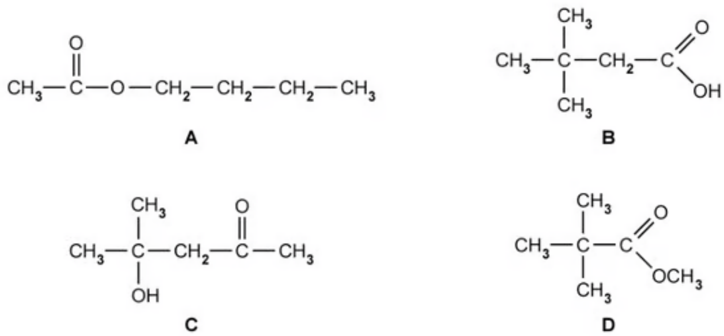


Fig. 6.4

The C-13 NMR spectrum of one of the four isomers of $C_6H_{12}O_2$ is shown in Fig. 6.5.

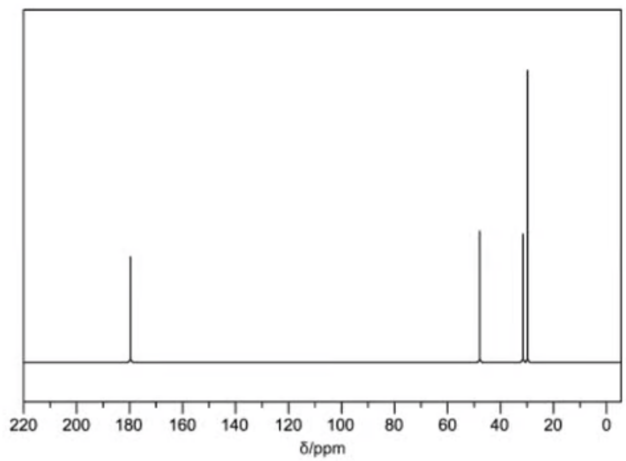


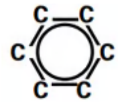
Fig. 6.5

The data in Table 6.3 should be used in answering this question.

Identify which of the four isomers, **A, B, C** or **D** of $C_6H_{12}O_2$ produced the C-13 NMR spectrum shown in Fig 6.5.

Table 6.3

Hybridisation of the carbon atom	Environment of carbon atom	Example	Chemical shift range δ/ppm
sp^3	alkyl	**C**H_3–, **C**H_2–, –**C**H<, >**C**<	0 – 50
sp^3	next to alkene / arene	–**C**–C=C, –**C**–Ar	25 – 50
sp^3	next to carbonyl / carboxyl	**C**–COR, **C**–O_2R	30 – 65
sp^3	next to halogen	**C**–X	30 – 60
sp^3	next to oxygen	**C**–O	50 – 70

sp^2	alkene or arene	>**C**=**C**<, 	110 – 160
sp^2	carboxyl	R–**C**OOH, R–**C**OOR	160 – 185
sp^2	carbonyl	R–**C**HO, R–**C**O–R	190 – 220
sp	nitrile	R–**C**≡N	100 – 125

[1 mark]

Question 7a

Ethane-1,2-diol, $C_2H_6O_2$, can be distinguished from ethanedioic acid, $C_2H_2O_4$, by a number of analytic techniques including MS, IR and NMR

Fig. 7.1 (spectrum **A**) and Fig. 7.2 (spectrum **B**) show the mass spectra of ethane-1,2-diol and ethanedioic acid.

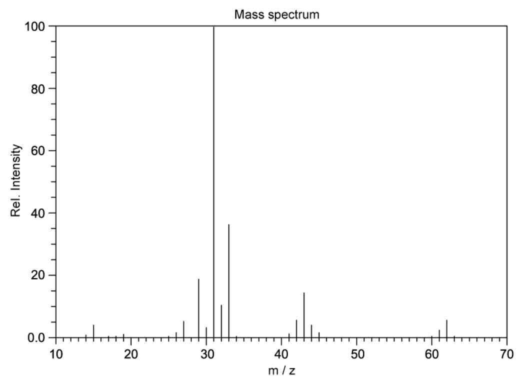


Fig. 7.1 (spectrum **A**)

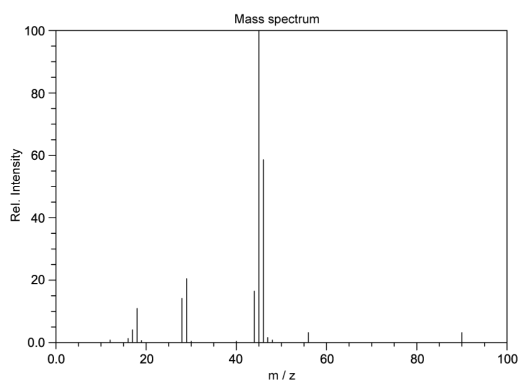


Fig. 7.2 (spectrum **B**)

Complete Table 7.1 to suggest which compound is responsible for each spectrum? Explain your answer.

Table 7.1

Spectrum	Organic compound	Explanation
A		
B		

[2 marks]

Question 7b

The IR spectra of ethane-1,2-diol, $C_2H_6O_2$, and ethanedioic acid dihydrate, $C_2H_2O_4.2H_2O$, are shown in Fig. 7.3 (Spectrum **C**) and Fig. 7.4 (Spectrum **D**).

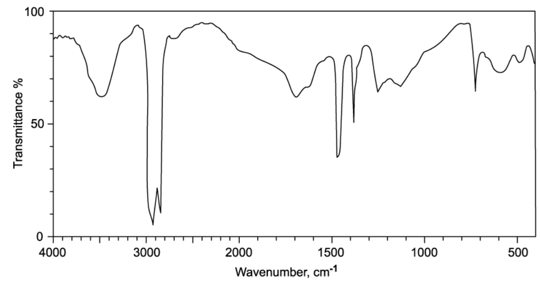


Fig. 7.3 (spectrum **C**)

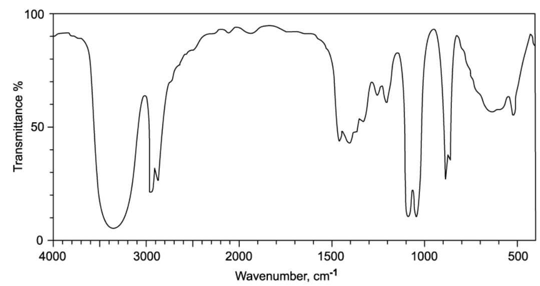


Fig. 7.4 (spectrum **D**)

The data in Table 7.3 should be used in answering this question.

Complete Table 7.2 to suggest which compound is responsible for each spectrum? Explain your answer.

Table 7.2

Spectrum	Organic compound	Explanation
C		
D		

Table 7.2

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumber) / cm^{-1}
C–O	hydroxy, ester	1040 – 1300
C=C	aromatic compound, alkene	1500 – 1680

C=O	amide carbonyl, carboxyl ester	1640 – 1690 1670 – 1740 1710 – 1750
C≡N	nitrile	2200 – 2250
C–H	alkane	2850 – 2950
N–H	amine, amide	3300 – 3500
O–H	carboxyl hydroxy	2500 – 3000 3200 – 3600

[2 marks]

Question 7c

The proton NMR spectrum of ethane-1,2-diol is shown in Fig. 7.5.

Describe and explain the splitting patterns of the spectrum.

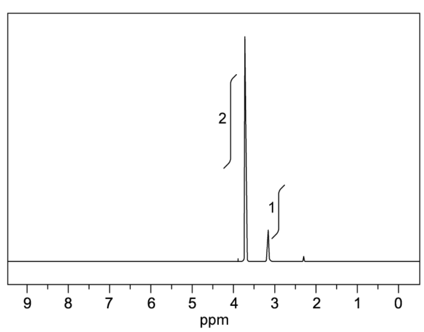


Fig. 7.5

[3 marks]

Question 7d

Suggest the number of proton NMR peaks and splitting pattern for ethanedioic acid.

[2 marks]

Model Answers: Medium

1a

a)

i) This method would **not** work because:

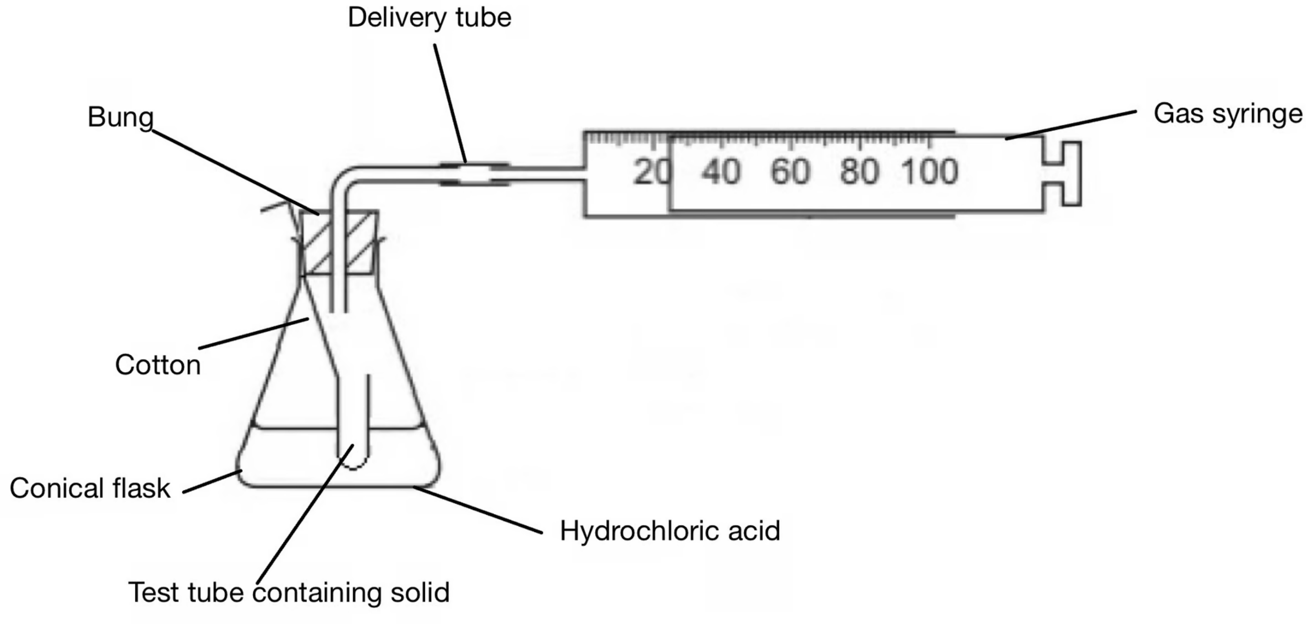
- $CuCO_3$ and $Cu(OH)_2$ both react (with HCl); [1 mark]

ii) You would accurately prepare 250.0 cm^3 of 0.500 mol dm^{-3} hydrochloric acid from the 10.0 mol dm^{-3} HCl provided

- (Transfer) 12.5(0) cm^3 of (10.0 mol dm^{-3}) HCl; [1 mark]
- Using a burette / graduated pipette; [1 mark]
- Add to a 250 cm^3 volumetric flask **AND** make to mark with distilled water; [1 mark]

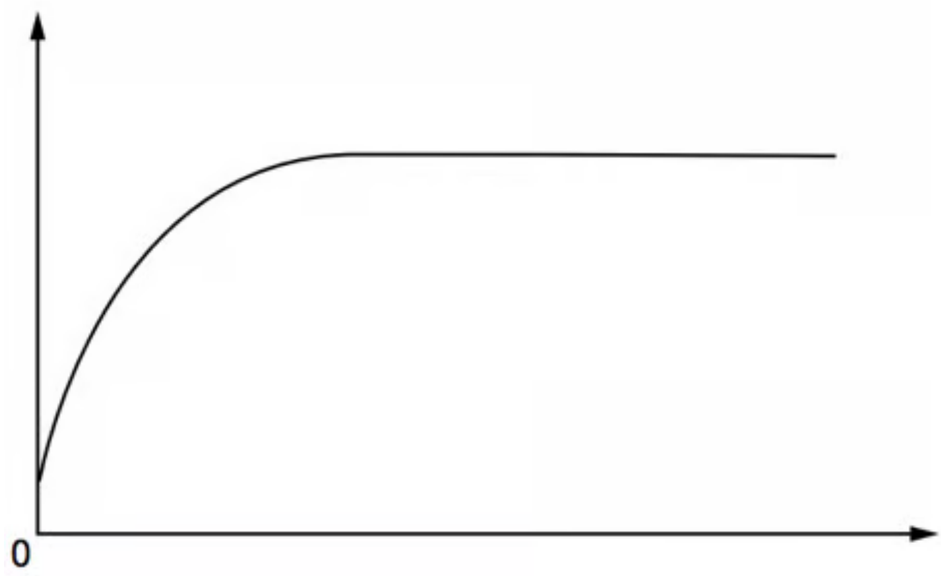
iii) A diagram to show how you would set up the apparatus and chemicals to measure the total volume of gas produced in this reaction should include:

- Suitable apparatus for production of CO_2 without initial gas loss e.g. divided conical flask or solid suspended in small test tube (using cotton) above acid; [1 mark]
- Suitable means of measuring CO_2 produced e.g. gas syringe or inverted measuring cylinder above water; [1 mark]



iv) A graph on the axes to show how the volume of gas produced would change during your experiment with the independent variable on the x-axis is:

- Correct labels on axes y-axis: volume (of gas)
  AND
  x-axis: time or t; [1 mark]
- Curved line (from origin) to reach a plateau; [1 mark]



v) To calculate the minimum volume, in cm^3, of 0.500 mol dm^{-3} HCl that is needed to completely react with this sample if the student is correct:

- Mol of $CuCO_3$ = 0.494 ÷ 123.5 = 4.00 × 10^{-3} mol; [1 mark]

- Volume of HCl = $2 \times 4.00 \times 10^{-3} \div 0.500 = 0.0160\ dm^3 = 16.0\ cm^3$; [1 mark]

[Total: 10 marks]

- You will be asked to give reasons why a particular method would not be suitable and you must use the clues in the question to help you explain why
  - $Cu(OH)_2$ is also present in verdigris so this will also react with acid as it is a base giving salt and water
  - $CuCO_3$ will react to give a salt, carbon dioxide and water
- This does however mean that we can collect the gas formed
  - Make sure your diagrams work so there are no gaps between bungs where gas can escape or blockages
  - Also, label your diagram
- To calculate the volume of gas formed use:
  - Mol = $\frac{\text{mass (g)}}{M_r}$ and then volume (dm^3) = $\frac{\text{mol}}{\text{concentration}}$
  - Make sure you show your working!

1b

b)

i) The completed table 1.1 is:

titration	rough	1	2	3
final reading / cm^3	25.55	23.90	48.30	28.10
initial reading / cm^3	0.00	0.00	23.90	3.95
titre / cm^3	25.55	23.90	24.40	24.15

- All correct; [1 mark]

ii) The percentage uncertainty in titre 1 is:

- $\frac{[(0.05 \times 2)]}{23.90} \times 100 = 0.42\ \%$; [1 mark]

iii) To calculate the percentage by mass of $Cu_3(CO_3)_2(OH)_2$ in the sample of azurite using the student's value of $24.15\ cm^3$ of $0.400\ mol\ dm^{-3}$ sulfuric acid:

- Moles of $H_2SO_4 = 0.40 \times \frac{24.15}{1000} = 9.66 \times 10^{-3}$ mol ; [1 mark]
- Mass of $Cu_3(CO_3)_2(OH)_2 = \frac{344.5 \times 9.66 \times 10^{-3}}{3} = 1.11$ g ; [1 mark]
- % by mass $= \frac{1.11}{1.50} \times 100\% = 74.0\%$; [1 mark]

iv) **Two** possible problems with the student's titration experiment and improvements to it are:

- Problem 1 titres are not concordant / are too far apart / are 0.5(0) cm^3 apart / difference is too large; [1 mark]
- Improvement 1 repeat until (two) concordant titres have been achieved / two readings within 0.1(0) cm^3; [1 mark]
- Problem 2 colour change (of indicator) will be difficult to see; [1 mark]
- Improvement 2 use an alternative indicator / named indicator; [1 mark]

[Total: 9 marks]

- A common error when working out percentage uncertainty for a burette reading is to not double the uncertainty
- When calculating % uncertainty consider if you are taking two readings (final and initial) or just a single reading
  - If you have two readings and you are given the precision for just one of the readings (like 0.05 for a burette), use
    - $\frac{2 \times \text{precision}}{\text{value to calculate \% uncertainty}} \times 100$
- An indicator that changes to a different colour from one colour is always difficult to see the end point clearly, so switching indicator will help

2a

a)

i) The completed graph is:

mass of activated charcoal, m / g	initial concentration of blue dye / g dm^{-3}	final concentration of blue dye, [X] / g dm^{-3}	difference in concentration of blue dye, D / g dm^{-3}	$\frac{D}{m}$	$\log \frac{D}{m}$	log [X]
0.20	25.00	0.96	24.04	120.20	2.08	-0.02

0.25	25.00	0.69	24.31	97.24	1.99	−0.16
0.30	25.00	0.60	24.40	81.33	1.91	−0.22
0.35	25.00	0.41	24.59	70.26	1.85	−0.39
0.40	25.00	0.33	24.67	61.68	1.79	−0.48
0.45	25.00	0.27	24.73	54.96	1.74	−0.57
0.50	25.00	0.23	24.77	49.54	1.69	−0.64
0.55	25.00	0.20	24.80	45.09	1.65	−0.70
0.60	25.00	0.17	24.83	41.38	1.62	−0.77

- D data correct
  AND
  To 2 decimal places ; [1 mark]
- Log [X] data correct
  AND
  To 2 decimal places ; [1 mark]

ii) The dependent variable in this experiment is:

- Final concentration of blue dye / [X]; [1 mark]

iii) Increasing the mass of activated charcoal, m, on the amount of adsorption that occurs:

- As mass of charcoal increases the final concentration of the dye decreases; [1 mark]
- Greater surface area leads to higher adsorption; [1 mark]

[Total: 5 marks]

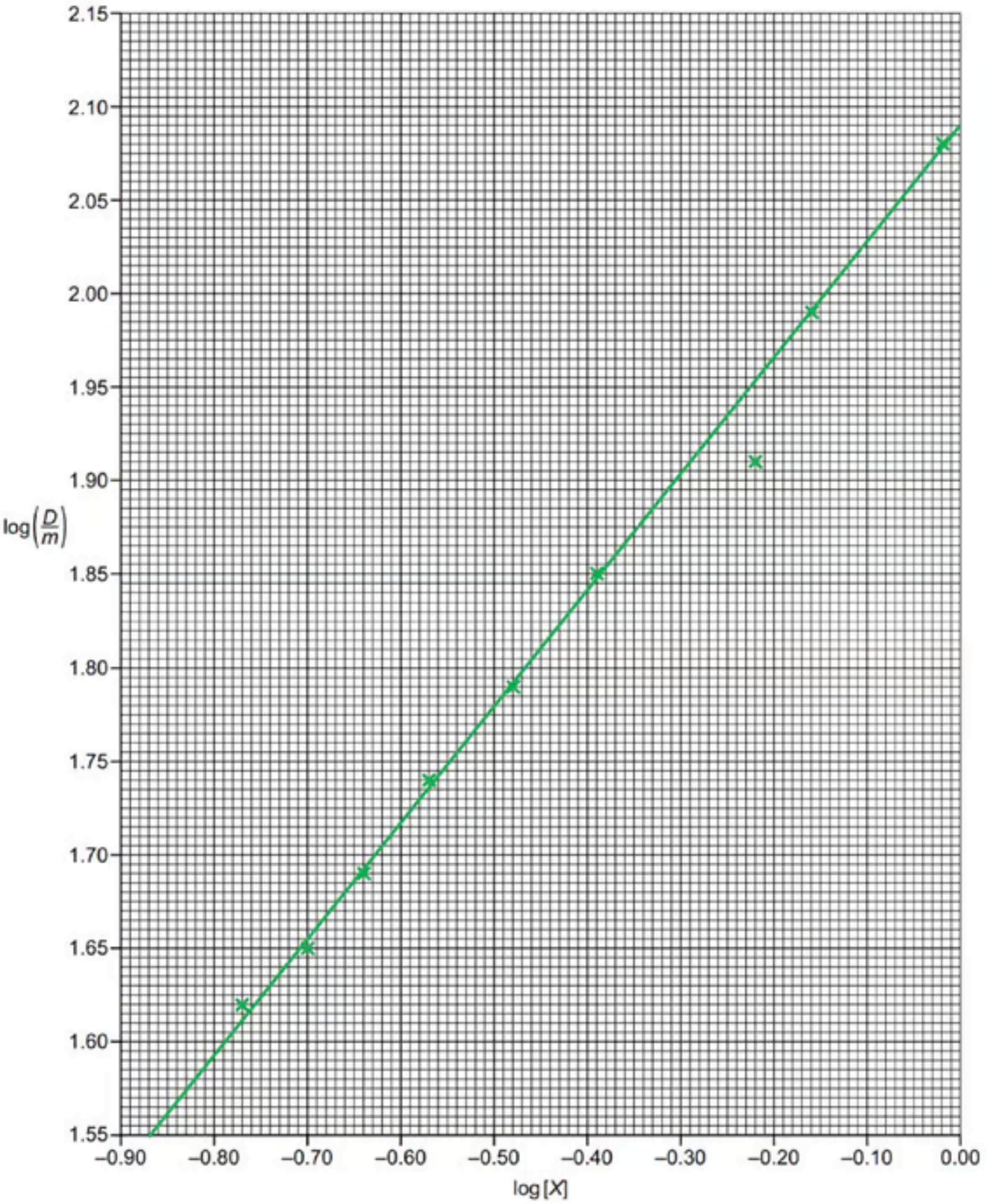
- Take note of how you must give your values
  - Here you have been asked for two decimal places
  - In some cases you could be asked for three significant figures
  - If you do not adhere to this you will lose the mark
- Take your time filling in the table and always check your answers
- If asked to fill in a table you are likely to be asked to plot a graph
- There are different types of variables within an experiment
  - The **independent variable**: the only variable that should be changed throughout an experiment, this is the one being investigated
  - The **controlled variables**: any other variables that may affect the results of

the experiment that need to be controlled
  - The **dependent variable**: the variable that we measure to judge the effect of changing the independent variable
- Therefore the dependent is the **final concentration of blue dye, [X]**
  - This is what the student wants to determine during the experiment
  - It is not what is calculated using the results

2b

b) The plotted points and line of best fit are:

- All nine points plotted correctly; [1 mark]
- (Straight) line of best fit drawn; [1 mark]



[Total: 2 marks]

- When plotting graphs and drawing lines of best fit
- Take your time
- Figure out the scale
  - Each small square on the x-axis is equivalent to a value of 0.01
  - Each small square on the y-axis is equivalent to a value of 0.005

    - This is clearly quite fiddly so use a ruler to line up the values on the axis to your points
- Plot your points with an 'x', otherwise, they will be hidden by the line of best fit
- For the line of best fit, it can be straight or a curve
    - You must use the most appropriate line
    - There should be an equal number of points above and below the line

2c

c) The most anomalous point on the graph and a reason why it may have happened is:

- Point (at –0.22, 1.91) identified
  AND

Any **one** of the following explaining the lack of adsorption:

- Not enough stirring; [1 mark]
- Mass of activated charcoal too low; [1 mark]
- Surface area not high enough / too low; [1 mark]
- Coagulation of charcoal; [1 mark]
- Bulkier particles used; [1 mark]
- Not left long enough; [1 mark]

[Total: 1 mark]

- Make sure you clearly circle your anomalous result
- As the result is lower than expected your explanation must link to this
- If the result is lower than expected, then the reaction can't have fully occurred so something must be preventing this from happening
    - Think about the method you have been given and think of a 'block' in the process that will cause the reading to be lower than expected

2d

d)

i) To determine the gradient of the line of best fit

- Coordinates read and recorded correctly (must not be points from the table unless on the line); [1 mark]

- Difference between the y coordinates must be at least half the range of the y values (0.3)
  AND
  Difference between the x coordinates must be at least half the range of the x values (0.45)
  AND
  Gradient determined (expected value = 0.625); [1 mark]

- Coagulation of charcoal; [1 mark]
- Bulkier particles used; [1 mark]
- Not left long enough; [1 mark]

[Total: 1 mark]

- Make sure you clearly circle your anomalous result
- As the result is lower than expected your explanation must link to this
- If the result is lower than expected, then the reaction can't have fully occurred so something must be preventing this from happening
  - Think about the method you have been given and think of a 'block' in the process that will cause the reading to be lower than expected

2d

d)

i) To determine the gradient of the line of best fit

- Coordinates read and recorded correctly (must not be points from the table unless on the line); [1 mark]
- Difference between the y coordinates must be at least half the range of the y values (0.3)
  AND
  Difference between the x coordinates must be at least half the range of the x values (0.45)
  AND
  Gradient determined (expected value = 0.625); [1 mark]

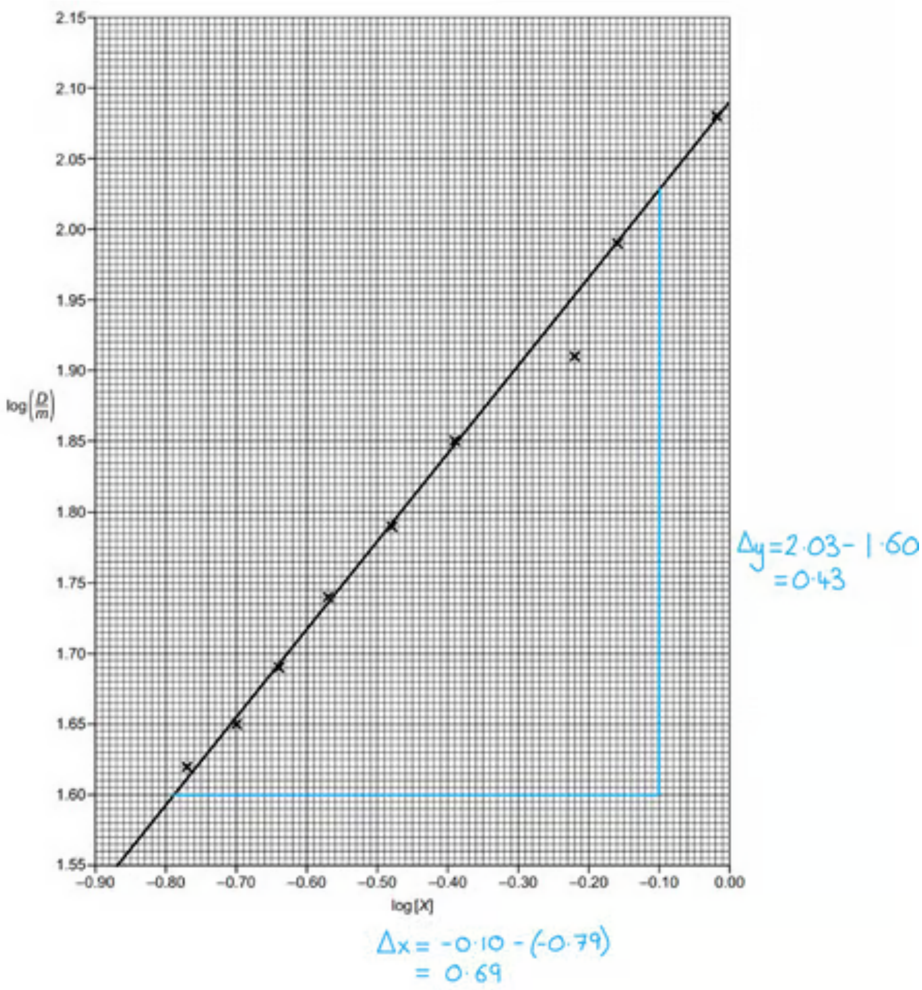
ii) To determine a value for A:

- Intercept on y-axis read and recorded correctly (expected value = 2.09 – 2.10); [1 mark]

[Total: 3 marks]

- To calculate the gradient, you must **not** use points from the table unless they are on your line of best fit
  - This is because you need to use your graph and line of best fit

- **Remember:** Gradient = $\frac{\Delta y}{\Delta x}$



- Therefore, your calculation is = $\frac{\mathbf{0.43}}{\mathbf{0.69}}$ = 0.623
- Since it is a straight-line graph, you can use y = mx + c to determine a value for 'A'
  - y = mx + c is the equation of a straight line
    - The variables x and y relate to coordinates on the line
    - m = gradient
    - c = y-intercept (when x = 0)
- With a little rearrangement, you apply the y = mx + c equation to the more complex $\log\left(\frac{D}{m}\right)$ equation
  - $\log\left(\frac{D}{m}\right) = A + b \log [X]$ can be rewritten as $\log\left(\frac{D}{m}\right) = b \log [X] + A$
  - Comparing this to y = mx + c:
    -

y	=	m	x	+	c

$\log\left(\frac{D}{m}\right)$	=	b	log [X]	+	A

- This shows that:
  - log [X] is the independent variable - which is reinforced by log [X] being the x-axis
  - $\log\left(\frac{D}{m}\right)$ is the dependent variable as it is determined by the value of x - which is further reinforced by $\log\left(\frac{D}{m}\right)$ being the y-axis
  - b is the gradient of the straight-line
  - A is the y-intercept of the straight line

3a

a) The difference between the mobile phases involved in column chromatography and gas-liquid chromatography is:

- The mobile phase in column chromatography is a liquid
  AND
  The mobile phase in gas-liquid chromatography is a gas; [1 mark]

[Total: 1 mark]

- This question could be made more challenging by using the word *eluent* instead of mobile phases
- Since you are told that column chromatography is similar to thin-layer chromatography, this question is basically asking what the difference between the mobile phases in thin-layer chromatography and gas-liquid chromatography is
  - **Remember:** Gas-liquid chromatography is another way of saying gas chromatography

3b

b) The compound with the greatest affinity for the solid phase is:

- Compound **C**
  AND
  (Because) Compound **C** has the greatest retention time; [1 mark]
- The longer the retention time; the greater the affinity for the solid phase; [1 mark]

[Total: 2 marks]

- Questions based on gas chromatograms are often linked to the affinity of a compound for the solid or gas phase
  - Examiners can also change the language to use mobile phase or stationary phase
- **Careful:** Gas chromatograms are often misinterpreted
  - **Remember:** Longer retention times mean that the compound is in the column for longer
  - This means that the compound must have a greater affinity for the solid / stationary phase

3c

c) The most abundant compound in the sample is:

- Compound **B**
  AND
  (Because) it has the highest / largest peak; [1 mark]

[Total: 1 mark]

- **Fig. 3.1** shows a typical example of a gas chromatogram
  - Examiners often ask you about the most abundant compound
  - To ask about the least abundant compound, an examiner would have to clearly show the area under each curve for you to deduce an answer
- **Remember:** The area under the peak is proportional to the abundance of that compound

3d

d) Gas chromatography could be used to identify the tanker as the source of crude oil pollution on the beach by:

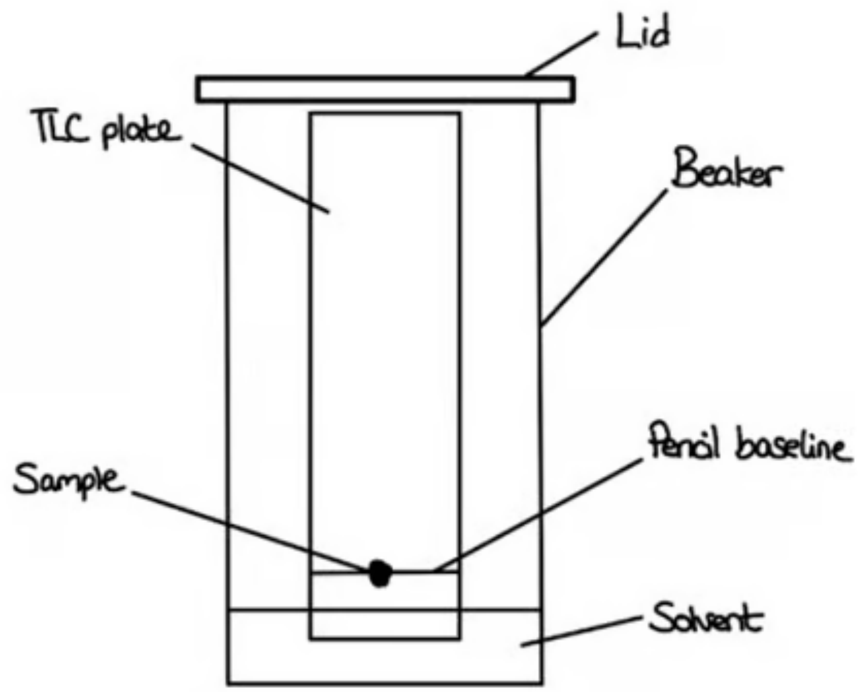
- Performing gas chromatography on a sample of oil from the tanker and a sample of oil from the beach; [1 mark]
- Comparing the chromatograms / results to see if they match; [1 mark]

[Total: 2 marks]

- The actual scenario of this question is a distraction and almost irrelevant
- From the actual question, you should be able to identify that you have two samples that need to be compared with one another
  - So, perform GC on one sample, perform GC on the other sample and compare the results to see if they match

4a

a) The labelled apparatus that would be used to identify the number of compounds in the mixture is:



- Correctly labelled diagram; [1 mark]
- Solvent level shown to be below the pencil baseline; [1 mark]

[Total: 2 marks]

- The question asks you for the labelled experimental set up to determine the number of compounds in the mixture by TLC
  - This means that you need to use a TLC plate and include a lid in your diagram
  - The TLC plate should have a pencil baseline with the sample spotted on it
  - The solvent **must** be below the pencil line - this is usually a specific point that examiners are looking for

4b

b) The steps that the student needs to perform to determine the identity of the compounds are:

- Measure the distance travelled by the solvent
  OR
  Measure the solvent front; [1 mark]
- Measure the distance travelled by each compound / component / spot; [1 mark]
- Calculate the R_f value using the equation $R_f = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$; [1 mark]
- Compare the R_f value of each compound / component / spot with data table values (to identify the compound / component / spot); [1 mark]

[Total: 4 marks]

- For this question, you need to describe how to proceed from a chromatogram to an R_f value
  - State the two distances that are required and the equation for R_f values
  - After that, the student is using R_f values to determine the identities of the compounds in the mixture so they will have to compare their R_f values against known R_f data

4c

c) An improved method to locate the centre of each spot is:

- Measure the distance to the top of each spot
  AND
  Measure the distance to the bottom of each spot; [1 mark]
- Add these values together and divide by 2
  OR
  Calculate the average of these distances
  OR
  Centre = $\frac{\text{Distance to top of the spot + distance to bottom of the spot}}{2}$; [1 mark]

[Total: 2 marks]

- The key to this question is deducing how to locate the centre of the spot
- Spots on a chromatogram can range from 2 mm to 40 mm on a TLC plate, which makes locating the centre difficult
- You can easily measure the distance to the top and bottom of the spot
- Then you calculate the average distance to get the centre

4d

d) To calculate the R_f values for both compounds in the chromatogram:

$$R_f \text{ value} = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$$

- Benzaldehyde = $\frac{3.2}{5}$ = 0.64; [1 mark]
- Benzyl alcohol = $\frac{1.5}{5}$ = 0.3; [1 mark]

[Total: 2 marks]

- To calculate R_f value or the Retention Factor the equation is
  - $R_f \text{ value} = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$
- Substances that are very soluble in the mobile phase will have less retention, due to weak adsorption on the stationary phase
  - R_f will be close to 1
- Substances that are not very soluble in the mobile phase will have large retention, due to strong adsorption with the stationary phase
  - R_f will be close to 0

4e

e) The maximum R_f value of a compound cannot exceed 1 because:

- R_f values are a ratio of the distance a compound has travelled compared to the distance travelled by the solvent; [1 mark]
- An R_f value of 1 implies / suggests that the compound is fully soluble in the solvent and does not interact with the stationary phase; [1 mark]

[Total: 2 marks]

- The challenge of this question is working out how to word your answer

- You might automatically think because it can't exceed 1 or because it's a fraction / decimal / ratio
- To answer this question
  - You need to apply that initial idea (especially if you thought ratio) to say what Rf is a ratio of
    - To be able to do this you need the equation as this tells you the ratio

$$R_f = \frac{\text{Distance travelled by compound}}{\text{Distance travelled by solvent}}$$

- You should also say what the maximum R_f value tells you about a compound
  - **Remember:** Higher R_f values mean:
    - The compound is more soluble in the solvent / has a greater affinity for the mobile phase
    - The compound has a lower affinity for the stationary phase

5a

a)

i) The structural formula of the standard reference chemical used in ^{1}H NMR is:

- $Si(CH_3)_4$; [1 mark]

ii) Tetramethylsilane is used as the standard reference chemical because:

Any **two** of the following:

- It is volatile / has a low boiling point ; [1 mark]
- It is non-toxic; [1 mark]
- It is inert ; [1 mark]
- It produces one single peak / signal; [1 mark]
- The peak it produces is far to the right / far away from other peaks; [1 mark]

[Total: 3 marks]

- The question is asking you to suggest the structural formula
  - **Remember:** Tetramethylsilane (TMS) is a silicon atom surrounded by four methyl groups
  - Therefore, its structural formula is $Si(CH_3)_4$
- TMS is used as the standard reference sample against which all other chemical shifts are measured

- It has a value of 0 ppm
- It will appear on the furthest right of an NMR spectrum
- It does not interfere with the other peaks
- It acts as a reference point to which the NMR spectrometer is tuned

5b

- It is volatile / has a low boiling point ; [1 mark]
- It is non-toxic; [1 mark]
- It is inert ; [1 mark]
- It produces one single peak / signal; [1 mark]
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[Total: 3 marks]

- The question is asking you to suggest the structural formula
  - **Remember:** Tetramethylsilane (TMS) is a silicon atom surrounded by four methyl groups
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  - It has a value of 0 ppm
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  - It does not interfere with the other peaks
  - It acts as a reference point to which the NMR spectrometer is tuned

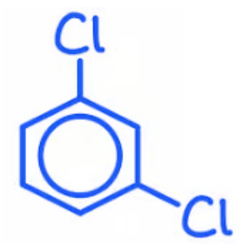
5b

b) The number of peaks in the compound 1,3-dichlorobenzene is;

- 4 / Four; [1 mark]

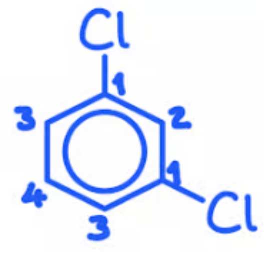
[Total: 1 mark]

- The best way to determine the number of C-13 peaks / environments for 1,3-dichlorobenzene is to draw the displayed formula:



- You can then determine the number of peaks by looking at the different positions of the carbon atoms within the molecule:
  - Carbon atoms that are bonded to different atoms or groups of atoms have different environments and will have different chemical shifts

    - If two carbon atoms are positioned symmetrically within a molecule, then they are equivalent and have the same environment
    - They will absorb radiation at the same chemical shift and contribute to the same peak
- 1,3-dichlorobenzene has four different environments for the carbon atoms
    - It is much easier to see this once you have drawn out the structural formula of the molecule



5c

c)

i) The number of peaks on the ^{13}C NMR of ethylbenzene is:

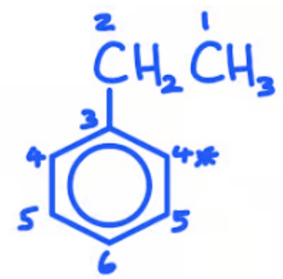
- 6 / six; [1 mark]

ii) The C-13 chemical shift range for the carbon with an * is:

- δ = 110 - 160 ppm; [1 mark]

[Total: 2 marks]

- The number of environments can be shown on the structural formula as shown:



- **Remember:** Look for carbon atoms that are bonded to different groups or positioned symmetrically within the molecule
    - Peaks 4 and 5 both show two sets of carbon atoms that are equivalent
    - This means that have the same chemical environment and will give the same chemical shift

- Looking at Table 5.1, the carbon with a * next to it will have a shift range of 110 - 160 ppm, for aromatic carbon atoms within ethylbenzene
    - The Kekule structure of benzene has been drawn here, which is unusual
    - If you are drawing benzene, while you will usually be allowed to use the Kekule structure, it would be better to use the delocalisation model using a circle to represent the delocalised ring of electrons

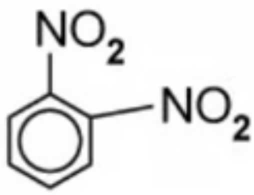
5d

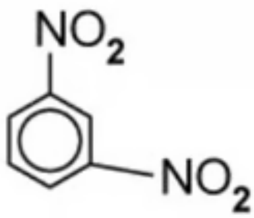
d)

i) The identity of the molecular ions that gives rise to the peak at *m / e* = 76 is:

- $C_6H_4^+$; [1 mark]

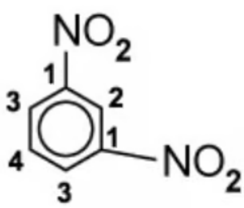
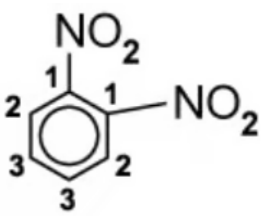
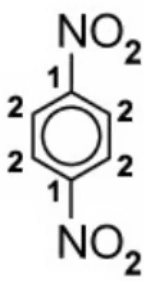
ii) The three possible isomers of compound **A** containing a benzene ring are below;





- Each correct structure; [1 mark]

iii) The labelled structures are:

Compound A = 1,3– dinitrobenzene	**Other isomer = 1,2– dinitrobenzene**	**Other isomer = 1,4 – dinitrobenzene**
 ; **[1 mark]**	 ; **[1 mark]**	 ; **[1 mark]**

[Total: 7 marks]

- The M_r of benzene = [(6 x 12.0) + (6 x 1.0)] = 78, so a peak at 76 would be caused by $C_6H_4^+$
  - Don't forget your positive charge on your ion
- It said in the question compound A has four peaks
  - 1,3–dinitrobenzene has four different carbon environments
  - The other two isomers have 3 and 2 respectively

6a

a)

i) $CDCl_3$ is used as a solvent instead of $CHCl_3$ because:

- It / $CDCl_3$ does not give a peak (on the spectrum)
  OR
  $CHCl_3$ does give a peak (on the spectrum); [1 mark]

ii) TMS is added to give the small peak at chemical shift δ = 0 because:

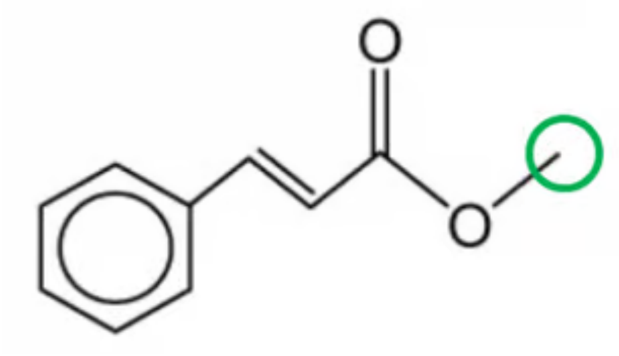
- Its chemical shifts can be compared /
  OR
  It is the reference / standard; [1 mark]

[Total: 2 marks]

- The use of TMS as a reference standard for calibration is a relatively common one-mark question

6b

b) The proton environment which causes the peak at chemical shift 3.8 ppm is:



- The -CH_3 group (in -OCH_3); [1 mark]
- It is a singlet caused by no adjacent / neighbouring hydrogen atoms; [1 mark]
- A peak area of 3 means there are 3 hydrogen atoms in this environment; [1 mark]

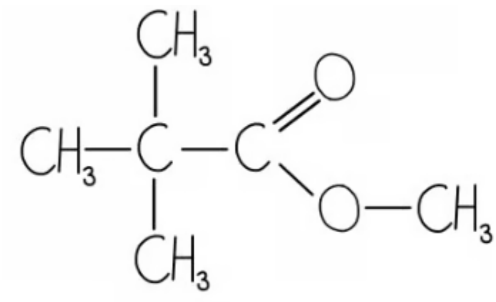
[Total: 3 marks]

- For splitting patterns remember the following;
  - Singlet (n + 1 = 1) - no H on adjacent atoms
  - Doublet (n + 1 = 2) - adjacent CH
  - Triplet (n + 1 = 3) - adjacent CH_2
  - Quartet (n + 1 = 4) - adjacent CH_3
- Table 6.1 shows that the chemical shift at δ 3.8 ppm can be an alkyl group next to an electronegative atom such as oxygen, C**H**$_3$–O, –C**H**$_2$–O
- From the spectrum, the peak at δ 3.8 ppm is a singlet
  - The only part of the compound that can produce a singlet is a -CH_3 group on the end of the molecule
  - This -CH_3 group is next to an electronegative oxygen atom which agrees with the proton NMR information in Table 6.1
- In your answer, you also have to reference that the peak area is 3, which tells you that there are three hydrogen atoms in this environment
- **Remember:** The ratio of the relative areas under each peak gives the ratio of the number of protons responsible for each peak

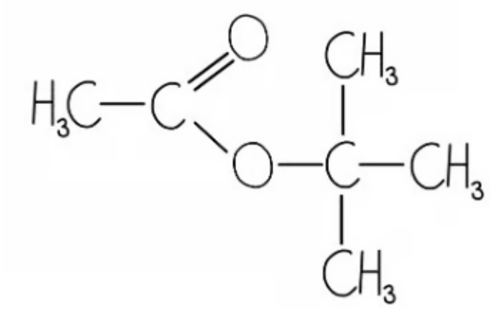
6c

c) The two esters with formula $C_6H_{12}O_2$ that each have only two peaks, both singlets, in their ^1H NMR spectra are:

- $(CH_3)_3CCOOCH_3$
  OR

 ; [1 mark]

- $CH_3COOC(CH_3)_3$
  OR

 ; [1 mark]

[Total: 2 marks]

- The first thing to notice in the question is that you are asked to draw two esters from the isomer $C_6H_{12}O_2$
- -COOC will feature in your drawings but make sure you try to focus on the fact that you need to draw the correct hydrogen environments as it says in the question the relative peak areas are 3:1 for both esters
  - This means that there are 3 CH_3 groups to 1 CH_3 group in the ester isomer and then in between them is the ester bond -COOC
- It is then a case of drawing one ester and then 'flipping' the three methyl groups to the other side of the ester bond
- **Remember:** Always double-check you have 4 bonds coming out of each carbon atom and not lose easy marks

6d

d)

i) The simplest ratio of protons in each environment is:

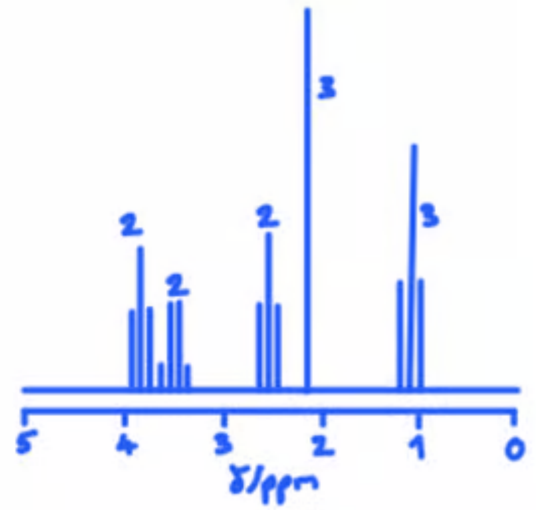
- 2:2:2:3:3 [1 mark]

ii) To describe and explain the splitting patterns of the peaks at δ = 3.5 and δ = 1.2:

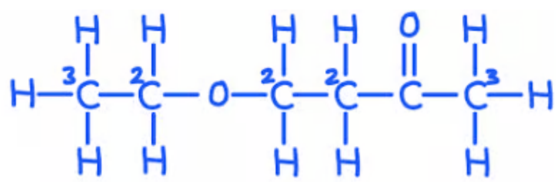
- δ 3.5 = is for a CH_2 group next to the -O-C**H**$_2$; [1 mark]
- δ 3.5 = is a quartet because of an adjacent CH_3 group; [1 mark]
- δ 1.2 = is for a C**H**$_3$-CH_2 ; [1 mark]
- δ 1.2 = is a triplet because of an adjacent CH_2 group; [1 mark]

[Total: 5 marks]

- **Remember:** The integration value tells you the ratio of the relative areas under each peak
- So an answer of 2:2:2:3:3 can be found as the **simplest whole number ratio from the integration value:**



- Write the corresponding ratios onto the spectrum so you can clearly see how the split patterns apply to the peaks
- You need to reference the **type of proton** the chemical shift value is showing as well as **explain what these split patterns are telling you**
- The isomer is drawn below, showing the relationship between the proton environments and the integration values:



6e

e) The isomer which produced the spectrum in Fig 6.5 is:

- B; [1 mark]

[Total: 1 mark]

- There are four peaks on the spectra which correspond to four different carbon environments
  - 30 ppm; C-CH_3 and there are three equivalent CH_3 groups
  - Just above 30 ppm; for the carbon atom with three CH_3 groups attached
  - 50 ppm; C-C=O for the CH_2 group

[Total: 1 mark]

- There are four peaks on the spectra which correspond to four different carbon environments
  - 30 ppm; C-CH_3 and there are three equivalent CH_3 groups
  - Just above 30 ppm; for the carbon atom with three CH_3 groups attached
  - 50 ppm; C-C=O for the CH_2 group
  - 180 ppm; RC=O for the COOH group

7a

a) The completed table assigning spectra **A** and **B** is:

Spectrum	Organic compound	Explanation
A	Ethane-1,2-diol	The highest *m/e* for spectrum **A** is 62 which matches the M_r of ethanediol,$C_2H_6O_2$
B	Ethanedioic acid	The highest *m/e* for spectrum **B** is 90 which matches the M_r of ethanedioic acid,$C_2H_2O_4$

- Spectrum **A**: Correct organic compound and explanation; [1 mark]
- Spectrum **B**: Correct organic compound and explanation; [1 mark]

[Total: 2 marks]

- Sometimes tiny peaks appear at M_r +1 which are due to isotopes, principally ^{13}C
- However, the amount of naturally occurring ^{13}C is around 1%, so it is unlikely to show up unless it is a very high-resolution mass spectrum

7b

b) The completed table assigning spectra **C** and **D** is:

Spectrum	Organic compound	Explanation
C	Ethanedioic acid	Spectrum **C** contains a peak at 1700–1750 cm^{-1} which matches the C=O in ethanedioic acid
		In spectrum D, there is a broad peak at 3200–

D	Ethane-1,2-diol	3600 cm^{-1} which is characteristic of O-H in alcohols

- Spectrum **C**: Correct organic compound and explanation; [1 mark]
- Spectrum **D**: Correct organic compound and explanation; [1 mark]

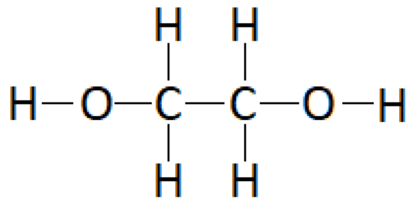
[Total: 2 marks]

- Ethanedioic acid is a hydrated solid acid, so there are O-H bonds from the water molecules as well as the carboxylic acid groups, making it a little hard to interpret the IR spectrum
- The best approach is to look for bonds that are unique to that molecule compared to ethane-1,2-diol

7c

c) The spectrum of ethane-1,2-diol shows:

- There are two peaks which indicate there are two unique ^1H /proton environments; [1 mark]
- The integration peaks show the ratio of the ^1H / proton environments is 2:1; [1 mark]
- This matches the structure of ethane-1,2-diol which has two identical -OH protons and four identical C-H protons; [1 mark]



[Total: 3 marks]

- You must distinguish between the number of protons and proton environments
- It would be incorrect to state there are two protons in one environment and one in another environment - it is a ratio, not the actual numbers

7d

d) The number of peaks and splitting pattern for ethanedioic acid:

- One peak / signal; [1 mark]
- There would be no splitting / a singlet; [1 mark]

[Total: 2 marks]

- Ethanedioic acid is a highly symmetrical molecule with only two protons in a chemically identical environment
- The single C–C is capable of rotating so the most sterically favourable position is